

2.1 Identity and Access Management

In this course, I will:

- Cover how AWS provides identity and access management (IAM) capabilities
- Explore how to create and manage IAM users and groups using the console, the CLI, and PowerShell
- Configure IAM policies and test user sign-in
- Configure IAM policies and roles, enable user MFA to harden user sign-ins, and use the IAM Identity Center console
- Cover how identity federation can provide centralized trusted identities

AWS Identity and Access Management (IAM)

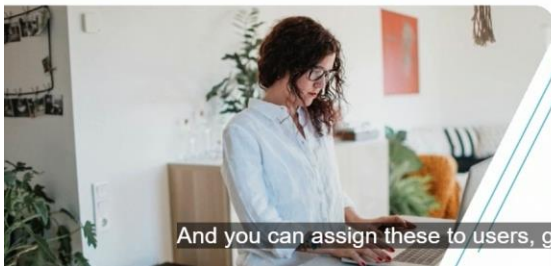
Management of:

1. Users
2. Groups
3. Roles

Authentication and Authorization

Authentication:

- Proof of identity
- Single factor: username, password
- Multi-factor: username, password, smartcard



And you can assign these to users, groups, or roles.



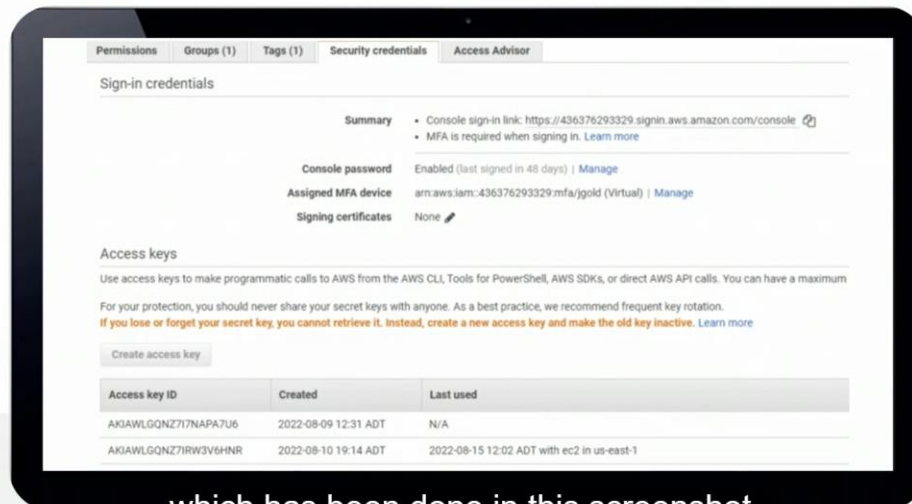
Authorization:

- Controlled access to apps and AWS resources
- Policies are collections of permissions
- Custom or built-in policies assigned to users, group, roles

Built-in AWS Policies

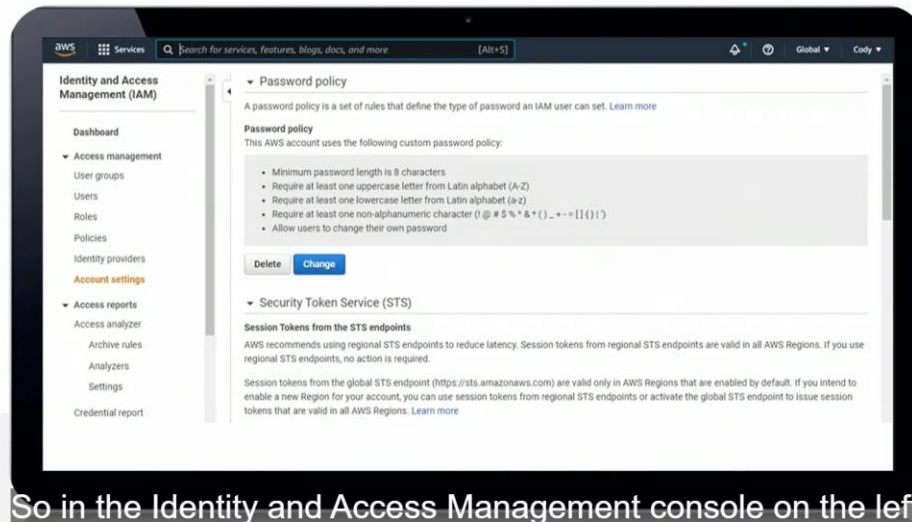
Policy name
AmazonEC2FullAccess
AmazonEC2ReadOnlyAccess
AmazonS3FullAccess
AmazonS3ReadOnlyAccess
AmazonRoute53DomainsFullAccess
AutoScalingFullAccess

AWS IAM User Security Credentials



which has been done in this screenshot.

AWS IAM Password Policy



So in the Identity and Access Management console on the left

Managing the IAM Users using GUI

Managing the IAM Users using CLI

Create user

The screenshot shows the AWS CloudShell interface with a terminal window. The terminal prompt is `[cloudshell-user@ip-10-0-3-39 ~]$`. The command entered is `aws iam create-user --user-name MBishop`. The output is a JSON object representing the created user.

```
[cloudshell-user@ip-10-0-3-39 ~]$ aws iam create-user --user-name MBishop
{
  "User": {
    "Path": "/",
    "UserName": "MBishop",
    "UserId": "AIDAWLGQNZ7I4IO2PPPYX",
    "Arn": "arn:aws:iam::436376293329:user/MBishop",
    "CreateDate": "2022-12-14T16:59:14+00:00"
  }
}
[cloudshell-user@ip-10-0-3-39 ~]$
```

set password for the user

The screenshot shows the AWS CloudShell interface with a terminal window. The terminal prompt is `[cloudshell-user@ip-10-0-3-39 ~]$`. The command entered is `aws iam create-login-profile --user-name MBishop --password Pa$$w0rdABC123 --password-reset-required`. The output is a JSON object representing the created login profile.

```
[cloudshell-user@ip-10-0-3-39 ~]$ aws iam create-login-profile --user-name MBishop --password Pa$$w0rdABC123 --password-reset-required
{
  "LoginProfile": {
    "UserName": "MBishop",
    "CreateDate": "2022-12-14T17:00:22+00:00",
    "PasswordResetRequired": true
  }
}
[cloudshell-user@ip-10-0-3-39 ~]$
```

Get user information

The screenshot shows the AWS CloudShell interface with a terminal window. The terminal prompt is `[cloudshell-user@ip-10-0-3-39 ~]$`. The command entered is `aws iam get-user --user-name MBishop`. The output is a JSON object representing the user information.

```
[cloudshell-user@ip-10-0-3-39 ~]$ aws iam get-user --user-name MBishop
{
  "User": {
    "Path": "/",
    "UserName": "MBishop",
    "UserId": "AIDAWLGQNZ7I4IO2PPPYX",
    "Arn": "arn:aws:iam::436376293329:user/MBishop",
    "CreateDate": "2022-12-14T16:59:14+00:00"
  }
}
[cloudshell-user@ip-10-0-3-39 ~]$
```

Reset password

```
[cloudshell-user@ip-10-0-3-39 ~]$ aws iam update-login-profile --user-name MBishop --password My\!User1ADifferentP@ssword
[cloudshell-user@ip-10-0-3-39 ~]$
```

Add user to a group

```
us-east-1
[cloudshell-user@ip-10-0-3-39 ~]$ aws iam add-user-to-group --user-name MBishop --group-name East-Developers
[cloudshell-user@ip-10-0-3-39 ~]$ aws iam get-user --user-name MBishop
{
  "User": {
    "Path": "/",
    "UserName": "MBishop",
    "UserId": "AIDAWLGQNZ7I4IO2PPPYX",
    "Arn": "arn:aws:iam::436376293329:user/MBishop",
    "CreateDate": "2022-12-14T16:59:14+00:00",
    "PasswordLastUsed": "2022-12-14T17:02:15+00:00"
  }
}
```

To check the group of the user

```
[cloudshell-user@ip-10-0-3-39 ~]$ aws iam list-groups-for-user --user-name MBishop
{
  "Groups": [
    {
      "Path": "/",
      "GroupName": "East-Developers",
      "GroupId": "AGPAWLGQNZ7IQ7BXJ7QVW",
      "Arn": "arn:aws:iam::436376293329:group/East-Developers",
      "CreateDate": "2022-12-14T17:04:13+00:00"
    }
  ]
}
```

Managing IAM Users Using PowerShell

```
Administrator: Windows PowerShell
PS C:\> get-awscredential -listprofiledetail

ProfileName StoreTypeName ProfileLocation
-----
admin1 NetSDKCredentialsFile
default NetSDKCredentialsFile

PS C:\>
```

```
Administrator: Windows PowerShell

PS C:\> initialize-awsdefaultconfiguration -profilename admin1
PS C:\> get-command *iamuser*

CommandType      Name                                     Version      Source
-----
Alias             Get-IAMUserPolicies                   4.1.196      AWSPowerShell
Alias             Get-IAMUsers                          4.1.196      AWSPowerShell
Cmdlet            Add-IAMUserTag                       4.1.196      AWSPowerShell
Cmdlet            Add-IAMUserToGroup                   4.1.196      AWSPowerShell
Cmdlet            Get-IAMUser                         4.1.196      AWSPowerShell
Cmdlet            Get-IAMUserList                     4.1.196      AWSPowerShell
Cmdlet            Get-IAMUserPolicy                   4.1.196      AWSPowerShell
Cmdlet            Get-IAMUserPolicyList               4.1.196      AWSPowerShell
Cmdlet            Get-IAMUserTagList                  4.1.196      AWSPowerShell
Cmdlet            New-IAMUser                         4.1.196      AWSPowerShell
Cmdlet            Register-IAMUserPolicy               4.1.196      AWSPowerShell
Cmdlet            Remove-IAMUser                     4.1.196      AWSPowerShell
Cmdlet            Remove-IAMUserFromGroup              4.1.196      AWSPowerShell
Cmdlet            Remove-IAMUserPermissionsBoundary    4.1.196      AWSPowerShell
Cmdlet            Remove-IAMUserPolicy                 4.1.196      AWSPowerShell
Cmdlet            Remove-IAMUserTag                   4.1.196      AWSPowerShell
Cmdlet            Set-IAMUserPermissionsBoundary       4.1.196      AWSPowerShell
Cmdlet            Unregister-IAMUserPolicy             4.1.196      AWSPowerShell
```

```
Select Administrator: Windows PowerShell

PS C:\> get-iamuser

Arn              : arn:aws:iam::436376293329:root
CreateDate       : 7/3/2018 6:07:58 PM
PasswordLastUsed : 12/14/2022 3:56:19 PM
Path             :
PermissionsBoundary :
Tags             : {}
UserId           : 436376293329
UserName         :
```

iamuser – root account

iamusers – all the users


```
Select Administrator: Windows PowerShell

PS C:\> get-iamusers

Arn      : arn:aws:iam::436376293329:user/JGold
CreateDate : 12/14/2022 4:08:57 PM
PasswordLastUsed : 12/14/2022 4:13:55 PM
Path     : /
PermissionsBoundary :
Tags     : {}
UserId   : AIDAWLGQNZ7I7XCLPZNVX
UserName : JGold

Arn      : arn:aws:iam::436376293329:user/MBishop
CreateDate : 12/14/2022 4:59:14 PM
PasswordLastUsed : 12/14/2022 5:02:15 PM
Path     : /
PermissionsBoundary :
Tags     : {}
UserId   : AIDAWLGQNZ7I4IO2PPPYX
UserName : MBishop
```

```
Select Administrator: Windows PowerShell

PS C:\> get-command *iamuser*

CommandType      Name                               Version      Source
-----
Alias            Get-IAMUserPolicies              4.1.196     AWSPowerShell
Alias            Get-IAMUsers                     4.1.196     AWSPowerShell
Cmdlet           Add-IAMUserTag                   4.1.196     AWSPowerShell
Cmdlet           Add-IAMUserToGroup               4.1.196     AWSPowerShell
Cmdlet           Get-IAMUser                      4.1.196     AWSPowerShell
Cmdlet           Get-IAMUserList                  4.1.196     AWSPowerShell
Cmdlet           Get-IAMUserPolicy                4.1.196     AWSPowerShell
Cmdlet           Get-IAMUserPolicyList            4.1.196     AWSPowerShell
Cmdlet           Get-IAMUserTagList               4.1.196     AWSPowerShell
Cmdlet           New-IAMUser                      4.1.196     AWSPowerShell
Cmdlet           Register-IAMUserPolicy            4.1.196     AWSPowerShell
Cmdlet           Remove-IAMUser                   4.1.196     AWSPowerShell
Cmdlet           Remove-IAMUserFromGroup          4.1.196     AWSPowerShell
Cmdlet           Remove-IAMUserPermissionsBoundary 4.1.196     AWSPowerShell
Cmdlet           Remove-IAMUserPolicy             4.1.196     AWSPowerShell
Cmdlet           Remove-IAMUserTag                4.1.196     AWSPowerShell
Cmdlet           Set-IAMUserPermissionsBoundary    4.1.196     AWSPowerShell
Cmdlet           Unregister-IAMUserPolicy          4.1.196     AWSPowerShell
Cmdlet           Update-IAMUser                   4.1.196     AWSPowerShell
Cmdlet           Write-IAMUserPolicy              4.1.196     AWSPowerShell

PS C:\> _
```

We can easily reset a user password in the CLI

```
Administrator: Windows PowerShell

PS C:\> new-iamuser -username AAdachi

Arn      : arn:aws:iam::436376293329:user/AAdachi
CreateDate : 12/14/2022 5:29:59 PM
PasswordLastUsed : 1/1/0001 12:00:00 AM
Path     : /
PermissionsBoundary :
Tags     : {}
UserId   : AIDAWLGQNZ7IYALFMPQNU
UserName : AAdachi

PS C:\> get-iamusers_
```

Select Administrator: Windows PowerShell

```
PS C:\> get-iamgroups
```

```
Arn      : arn:aws:iam::436376293329:group/East-Developers
CreateDate : 12/14/2022 5:04:13 PM
GroupId   : AGPAWLQONZ7IQ7BXJ7QVW
GroupName : East-Developers
Path      : /
```

```
PS C:\> _
```

Administrator: Windows PowerShell

```
PS C:\> add-iamusertogroup -username aadachi -groupname east-developers
```

```
PS C:\> _
```

Select Administrator: Windows PowerShell

```
PS C:\> get-iamgroup -groupname east-developers
```

Group	IsTruncated	Marker	Users

Amazon.IdentityManagement.Model.Group	False		{MBishop, AAdachi}

```
PS C:\> _
```


Administrator: Windows PowerShell

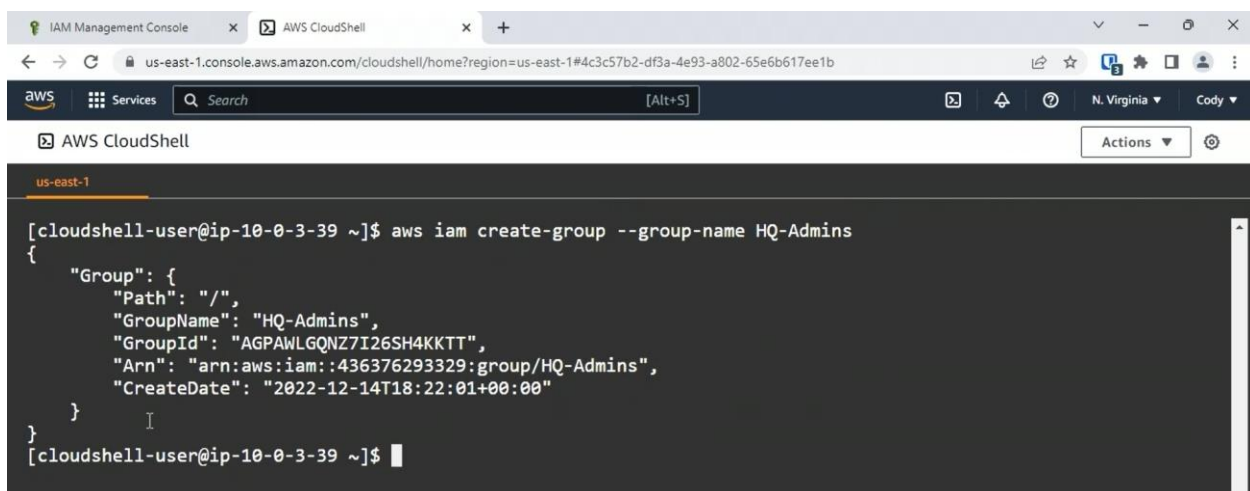
```
PS C:\> $console = @{  
>> UserName = 'AAdachi'  
>> Password = 'Pa$$w0rdABC123'  
>> PasswordResetRequired = $true  
>> }  
PS C:\> new-iamloginprofile @console
```

```
PS C:\> _
```

Managing IAM Groups Using the GUI

Managing IAM Groups Using the CLI

Aws iam help



```
us-east-1  
[cloudshell-user@ip-10-0-3-39 ~]$ aws iam create-group --group-name HQ-Admins  
{  
  "Group": {  
    "Path": "/",  
    "GroupName": "HQ-Admins",  
    "GroupId": "AGPAWL6QNZ7I26SH4KKTT",  
    "Arn": "arn:aws:iam::436376293329:group/HQ-Admins",  
    "CreateDate": "2022-12-14T18:22:01+00:00"  
  }  
}  
[cloudshell-user@ip-10-0-3-39 ~]$
```

aws iam list-groups – to get list of groups

```
aws
Services Search [Alt+S] N. Virginia Cody
AWS CloudShell Actions
us-east-1
[cloudshell-user@ip-10-0-3-39 ~]$ aws iam list-groups --query "Groups[].Groupname"
[]
[cloudshell-user@ip-10-0-3-39 ~]$ aws iam list-groups --query "Groups[].GroupName"
[
  "East-Developers",
  "HQ-Admins",
  "West-Developers"
]
[cloudshell-user@ip-10-0-3-39 ~]$
```

```
us-east-1
[cloudshell-user@ip-10-0-3-39 ~]$ aws iam get-group --group hq-admins
{
  "Users": [],
  "Group": {
    "Path": "/",
    "GroupName": "HQ-Admins",
    "GroupId": "AGPAWLQNZ7I26SH4KKTT",
    "Arn": "arn:aws:iam::436376293329:group/HQ-Admins",
    "CreateDate": "2022-12-14T18:22:01+00:00"
  }
}
[cloudshell-user@ip-10-0-3-39 ~]$
```

Add and remove user from group

```
[cloudshell-user@ip-10-0-3-39 ~]$ aws iam add-user-to-group --user-name JGold --group-name hq-admins
[cloudshell-user@ip-10-0-3-39 ~]$ aws iam remove-user-from-group --group-name hq-admins --user-name jgold
[cloudshell-user@ip-10-0-3-39 ~]$
```

Managing IAM Groups Using PowerShell

```
Administrator: Windows PowerShell
PS C:\> get-command *iamgroup*

CommandType      Name                                     Version        Source
-----
Alias             Get-IAMGroupPolicies                  4.1.196        AWSPowerShell
Alias             Get-IAMGroups                         4.1.196        AWSPowerShell
Cmdlet            Get-IAMGroup                         4.1.196        AWSPowerShell
Cmdlet            Get-IAMGroupForUser                  4.1.196        AWSPowerShell
Cmdlet            Get-IAMGroupList                    4.1.196        AWSPowerShell
Cmdlet            Get-IAMGroupPolicy                  4.1.196        AWSPowerShell
Cmdlet            Get-IAMGroupPolicyList              4.1.196        AWSPowerShell
Cmdlet            New-IAMGroup                        4.1.196        AWSPowerShell
Cmdlet            Register-IAMGroupPolicy              4.1.196        AWSPowerShell
Cmdlet            Remove-IAMGroup                    4.1.196        AWSPowerShell
Cmdlet            Remove-IAMGroupPolicy              4.1.196        AWSPowerShell
Cmdlet            Unregister-IAMGroupPolicy            4.1.196        AWSPowerShell
Cmdlet            Update-IAMGroup                    4.1.196        AWSPowerShell
Cmdlet            Write-IAMGroupPolicy                4.1.196        AWSPowerShell

PS C:\>
```

```
Administrator: Windows PowerShell
PS C:\> get-iamgroup -groupname east-developers

Group                                     IsTruncated Marker Users
-----
Amazon.IdentityManagement.Model.Group False          {MBishop, AAdachi}

PS C:\>
```

```
PS C:\> get-iamgroups

Arn      : arn:aws:iam::436376293329:group/East-Developers
CreateDate : 12/14/2022 5:04:13 PM
GroupId   : AGPAWLGQNZ7IQ7BX17QVW
GroupName : East-Developers
Path      : /

Arn      : arn:aws:iam::436376293329:group/HQ-Admins
CreateDate : 12/14/2022 6:22:01 PM
GroupId   : AGPAWLGQNZ7I26SH4KKTT
GroupName : HQ-Admins
Path      : /

Arn      : arn:aws:iam::436376293329:group/West-Developers
CreateDate : 12/14/2022 6:05:46 PM
GroupId   : AGPAWLGQNZ7I2DYA5AE4L
GroupName : West-Developers
Path      : /
```

```

Administrator: Windows PowerShell
PS C:\> get-iamgroups | select groupname

GroupName
-----
East-Developers
HQ-Admins
West-Developers

PS C:\> new-iamgroup -groupname MigrationTeam

Arn      : arn:aws:iam::436376293329:group/MigrationTeam
CreateDate : 12/15/2022 11:32:41 AM
GroupId   : AGPAWLGQNZ7IW33SVQKKZ
GroupName : MigrationTeam
Path      : /

PS C:\>

```

```

Administrator: Windows PowerShell
PS C:\> add-iamusertogroup -username mbishop -groupname MigrationTeam
PS C:\> get-iamgroup -groupname MigrationTeam

Group                                     IsTruncated Marker Users
-----
Amazon.IdentityManagement.Model.Group False          {MBishop}

PS C:\>

```

Attach policy

```

PS C:\> Register-iamgrouppolicy -groupname MigrationTeam -policyarn arn:aws:iam::aws:policy/AWSLambdaExecute
PS C:\> get-iamattachedgrouppolicies -groupname MigrationTeam

PolicyArn                                     PolicyName
-----
arn:aws:iam::aws:policy/AWSLambdaExecute AWSLambdaExecute

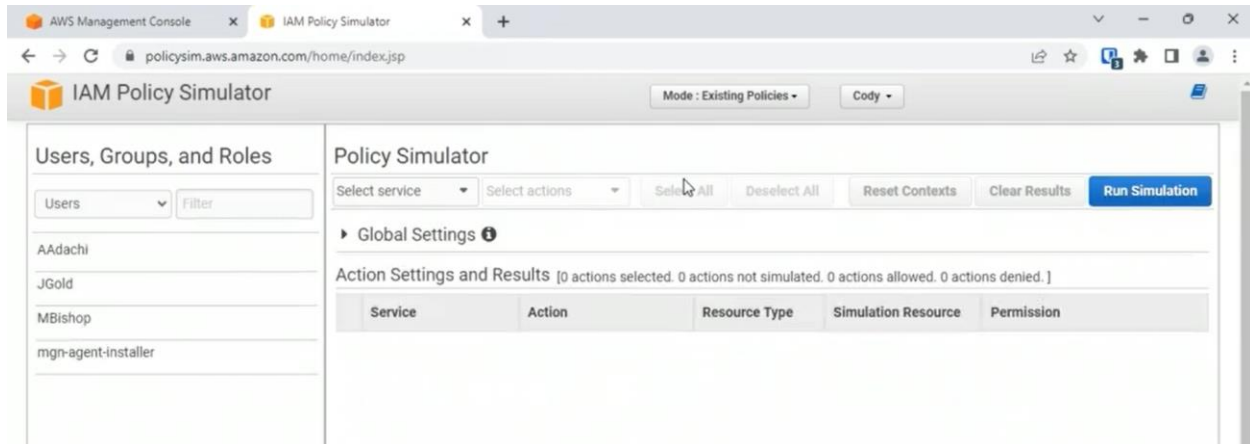
PS C:\>

```

Configuring IAM policies

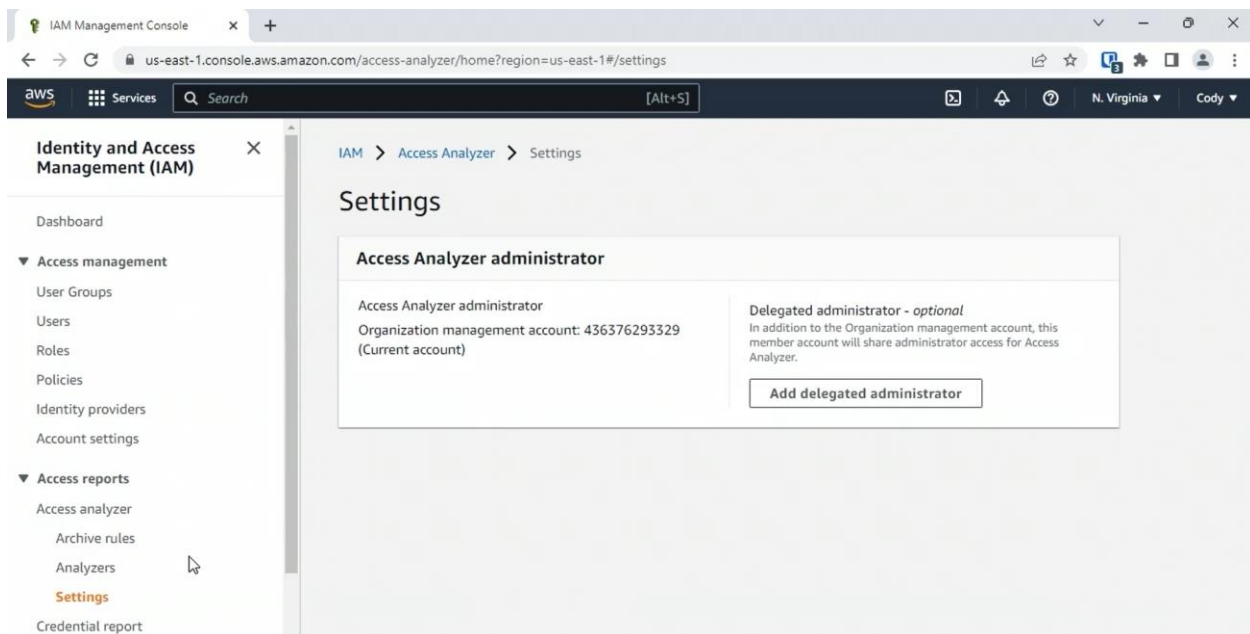
Configuring IAM Roles

Using the IAM Policy Simulator



Can select users, groups and roles from the dropdown

Using the IAM Access Analyzer



IAM Management Console

us-east-1.console.aws.amazon.com/access-analyzer/home?region=us-east-1#/findings

Identity and Access Management (IAM)

- Dashboard
- Access management
 - User Groups
 - Users
 - Roles
 - Policies
 - Identity providers
 - Account settings
- Access reports
 - Access analyzer
 - Archive rules
 - Analyzers
 - Settings

Analyzer

ConsoleAnalyzer-363867bc-93f2-490d-bc8f-d4cc19eaf003

Zone of trust: Current account (436376293329)

Active Archived Resolved All

Active findings

Account ID 436376293329

Filter active findings

External princ...	Condition	Shared throu...	Access le...	Upd...
All Principals	-	Bucket policy	Read	14 day...
Federated User arn:aws:iam::43...	-	-	Write	2 mon...
AWS Account	-	-	-	-

IAM Management Console

us-east-1.console.aws.amazon.com/access-analyzer/home?region=us-east-1#/findings/details/08996df8-10dd-4a3c-b81b-02bfc3e7396e

Identity and Access Management (IAM)

- Dashboard
- Access management
 - User Groups
 - Users
 - Roles
 - Policies
 - Identity providers
 - Account settings
- Access reports
 - Access analyzer
 - Archive rules
 - Analyzers
 - Settings

Resource

arn:aws:s3:::website80091

External principal

All Principals

Condition

-

Access level

Read

- s3:GetObject

Resource owner account

436376293329

Next steps

Intended access

If the access is intended, such as access necessary for business processes, you can archive the finding. This lets you focus on findings that are related to potential security risks. When you archive a finding, it's removed from Active findings and the status changes to Archived.

Archive

To automatically archive similar findings, create an archive rule.

Not intended

If the access isn't intended, it indicates a potential security risk. Use the console for the service associated with the resource to modify or remove the policy that grants the unintended access. To confirm that your change removed the access, choose **Rescan**. If the access is removed, the status changes to Resolved.

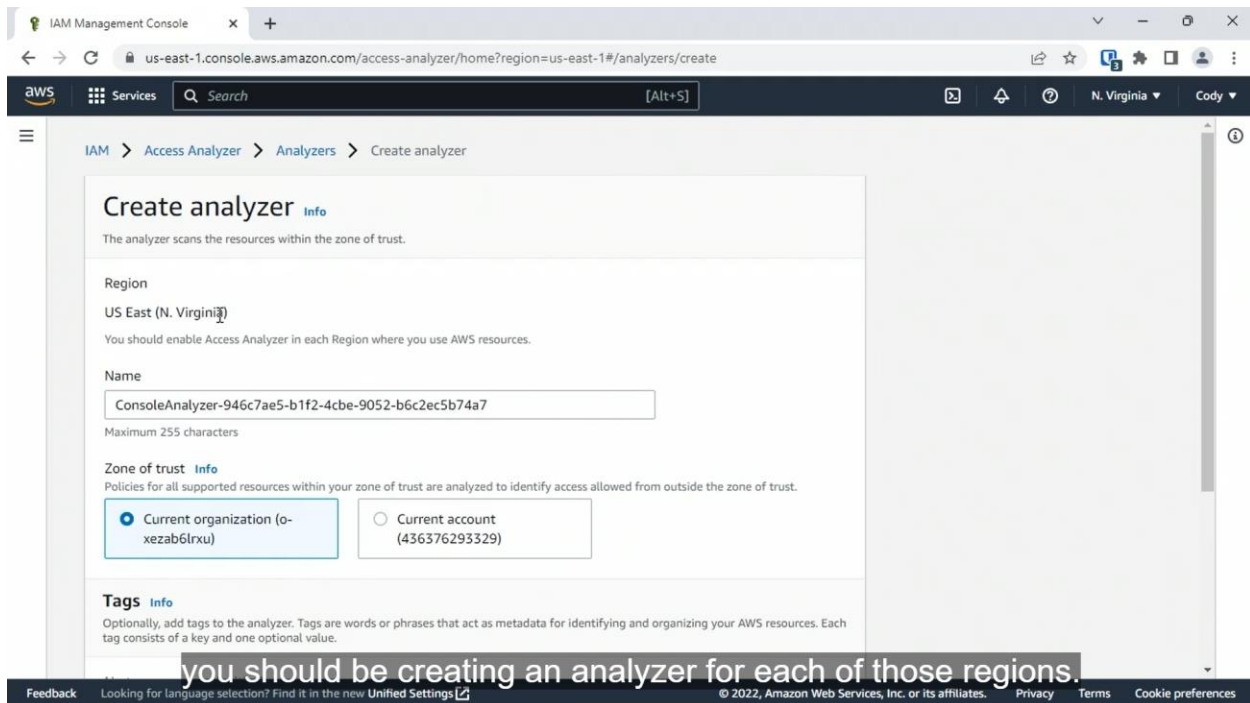
Go to S3 console

arn:aws:s3:::website80091

to rectify the situation,

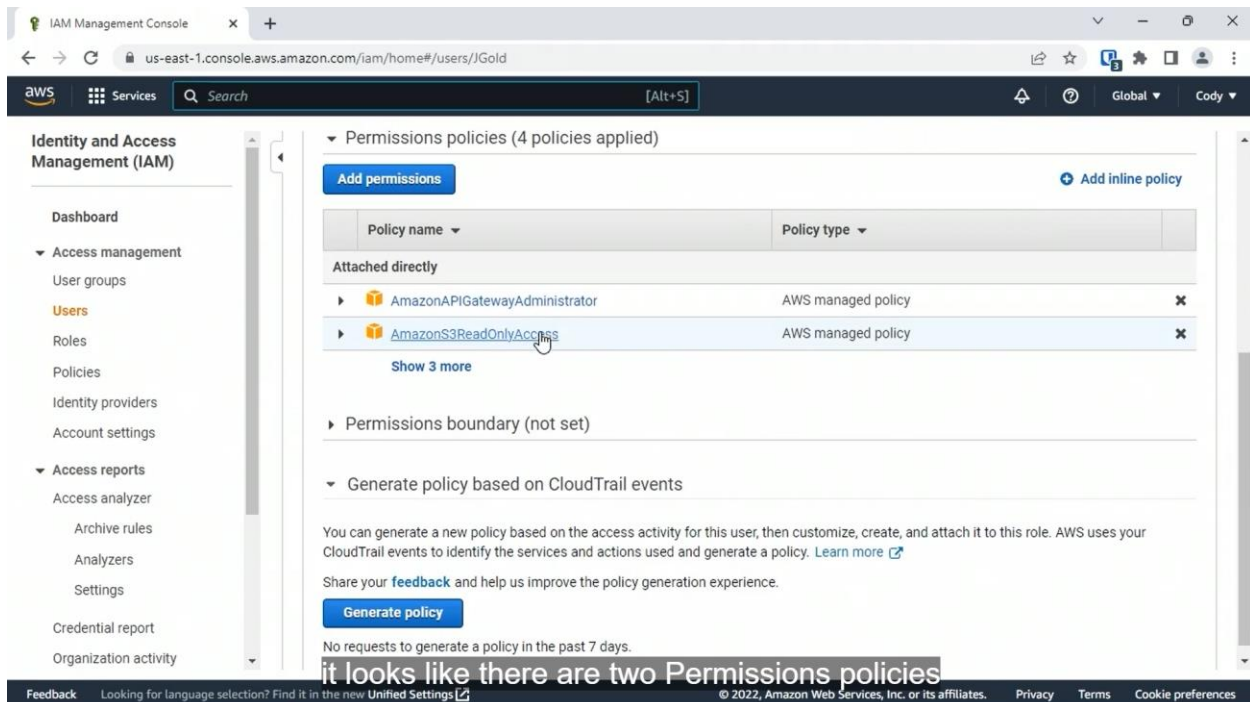
Feedback Looking for language selection? Find it in the new Unified Settings

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Working with IAM Permissions Boundaries

Will find permission boundaries under the individual user or roles and not in the user groups



IAM Management Console x IAM Policy Simulator x +

us-east-1.console.aws.amazon.com/iam/home#/users/putScopePermissions?id=JGold&type=user&action=add&step=putScope

aws Services Search [Alt+S]

Set the permissions boundary on JGold

Setting the permissions boundary is an advanced feature
The permissions boundary can restrict the permissions currently allowed for this user. [Learn more](#)

Select a policy to set as the permissions boundary

Create policy

Filter policies Q s3 Showing 20 results

	Policy name	Type	Used as
☐	AmazonS3OutpostsReadOnlyAccess	AWS managed	None
☐	AmazonS3OutpostsReadOnlyAccess	AWS managed	None
☒	AmazonS3ReadOnlyAccess	AWS managed	Permissions policy (2)
☐	AWSBackupServiceRolePolicyForS3Backup	AWS managed	None

configuration level;

Cancel Set boundary

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Enabling MFA for an IAM User

IAM Management Console x +

us-east-1.console.aws.amazon.com/iam/home#/users/MBishop?section=security_credentials

aws Services Search [Alt+S]

Identity and Access Management (IAM)

Dashboard

Access management

User groups

Users

Roles

Policies

Identity providers

Account settings

Access reports

Access analyzer

Archive rules

Analyzers

Settings

Credential report

Organization activity

Path /

Creation time 2022-12-14 12:59 AST

Permissions Groups (2) Tags **Security credentials** Access Advisor

Sign-in credentials

Summary • Console sign-in link: <https://436376293329.signin.aws.amazon.com/console>

Console password Enabled (last signed in Yesterday) | [Manage](#)

Signing certificates None

Multi-factor authentication (MFA)

Use MFA to increase the security of your AWS environment. Signing in with MFA requires an authentication code from an MFA device. You can assign a maximum of 8 MFA devices. [Learn more](#)

[Manage](#) [Assign MFA device](#)

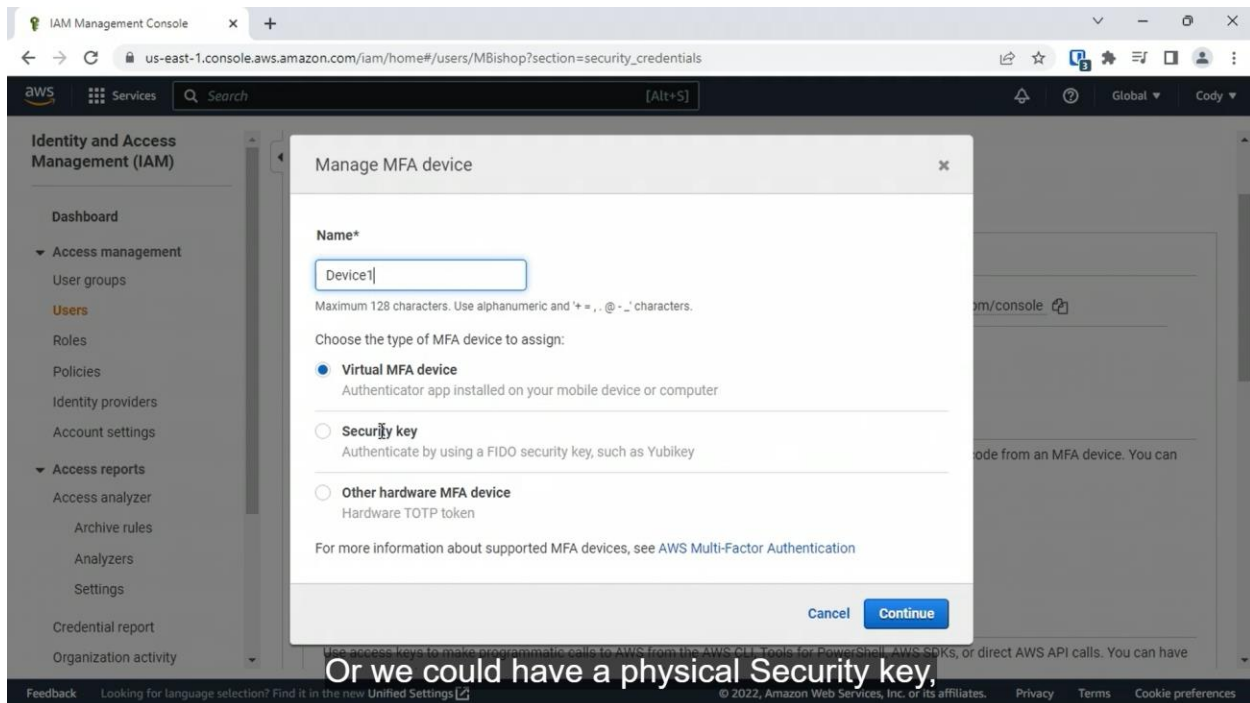
No MFA assigned.

Access keys

Use access keys to make programmatic calls to AWS from the AWS CLI, Tools for PowerShell, AWS SDKs, or direct AWS API calls. You can have

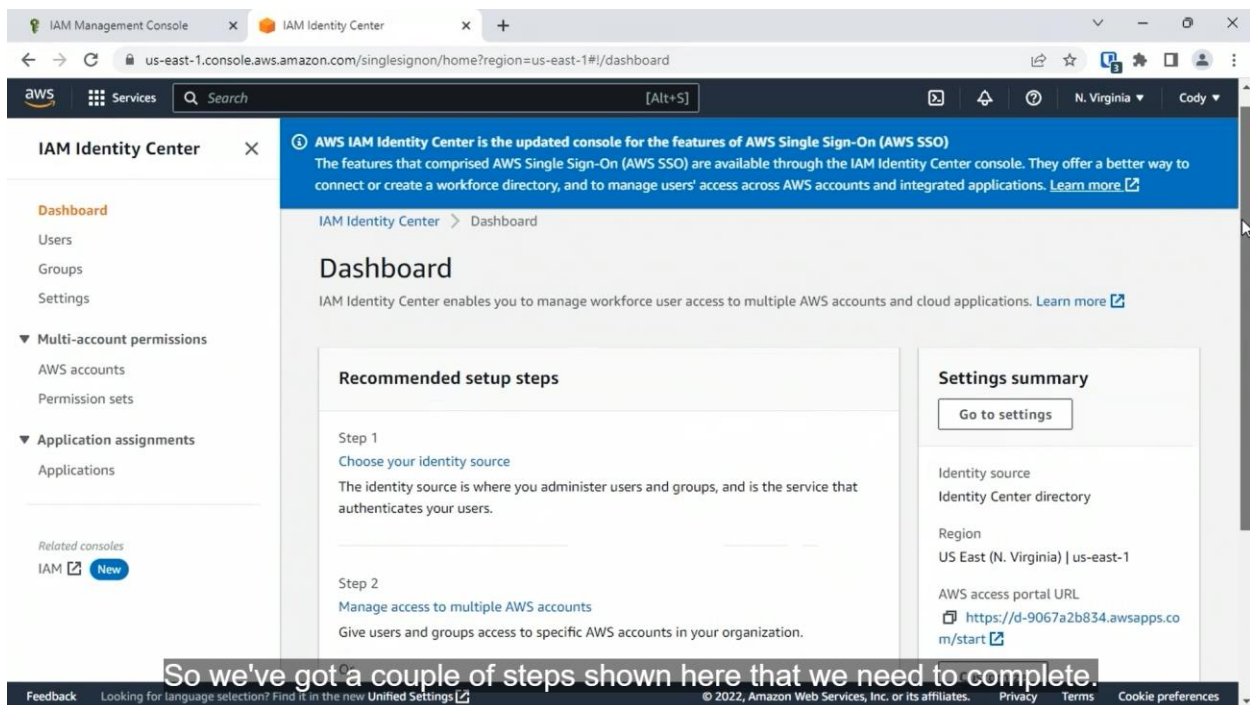
under the Multi-factor authentication

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Using the IAM Identity Center Console

First enable the identity center



AWS Federated Access

Identity federation

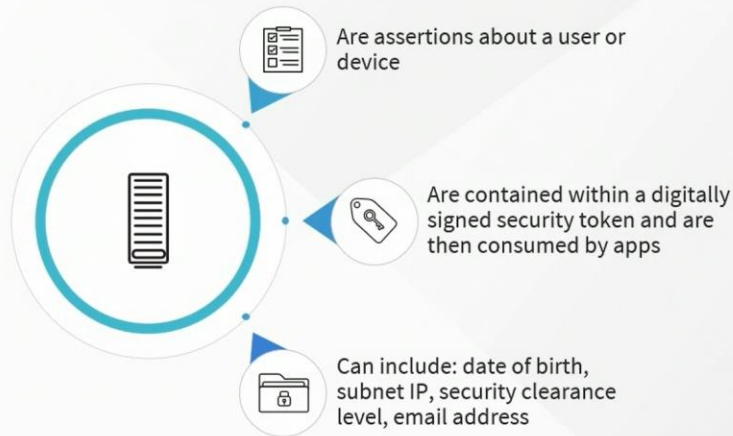
- Uses a centralized trusted authenticator
- Authentication capabilities are removed from apps
- Apps rely upon the trusted authenticator
- A trusted authenticator can span organizations

Identity and Resource Providers



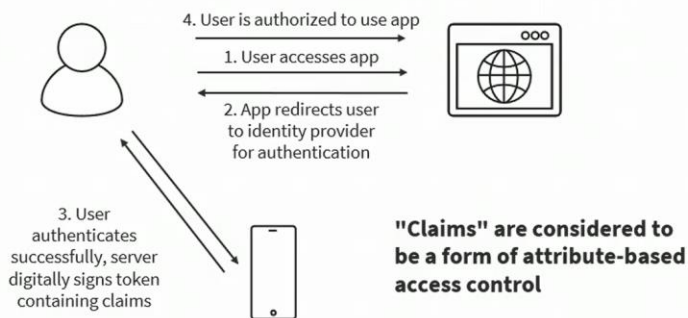
and sometimes it's even called a service provider, an SP.

Identity Federation Claims



to determine what level of access

Identity Federation

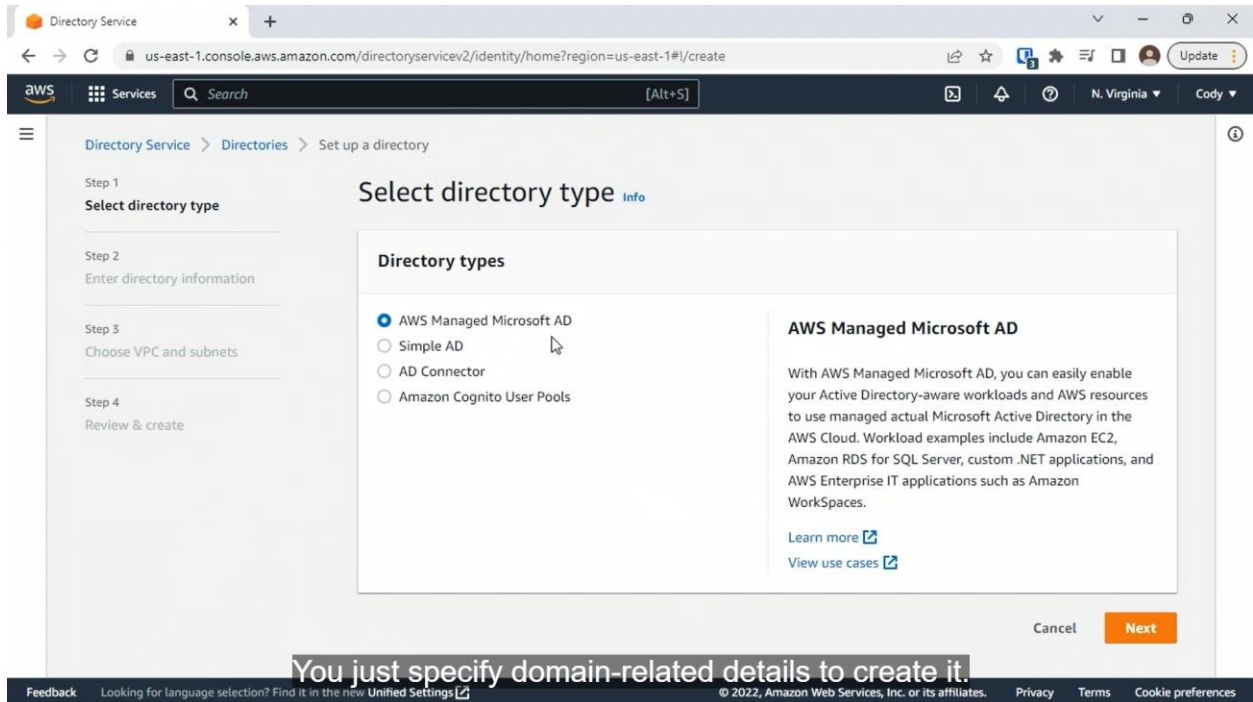


but this is one possibility.

AWS Identity Solutions

- AWS Single Sign-On (SSO) > AWS IAM Identity Center
- AWS Security Token Service: temporary for like web applications

Deploying AWS Managed Active Directory



Select directory type [Info](#)

Directory types

- ☐ AWS Managed Microsoft AD
- ☒ Simple AD
- ☐ AD Connector
- ☐ Amazon Cognito User Pools

Simple AD

Simple AD is a standalone managed directory that is powered by a Linux-Samba Active Directory-compatible server.

[Learn more](#)

Cancel

Next

Select directory type [Info](#)

Directory types

- ☐ AWS Managed Microsoft AD
- ☐ Simple AD
- ☒ AD Connector
- ☐ Amazon Cognito User Pools

AD Connector

AD Connector is a proxy for redirecting directory requests to your existing Microsoft Active Directory without caching any information in the cloud. AD Connector comes in two sizes, small and large. A small AD Connector is designed for small organizations and is intended to handle a low number of operations per second. A large AD Connector is designed for large organizations and is intended to handle a moderate to high number of operations per second.

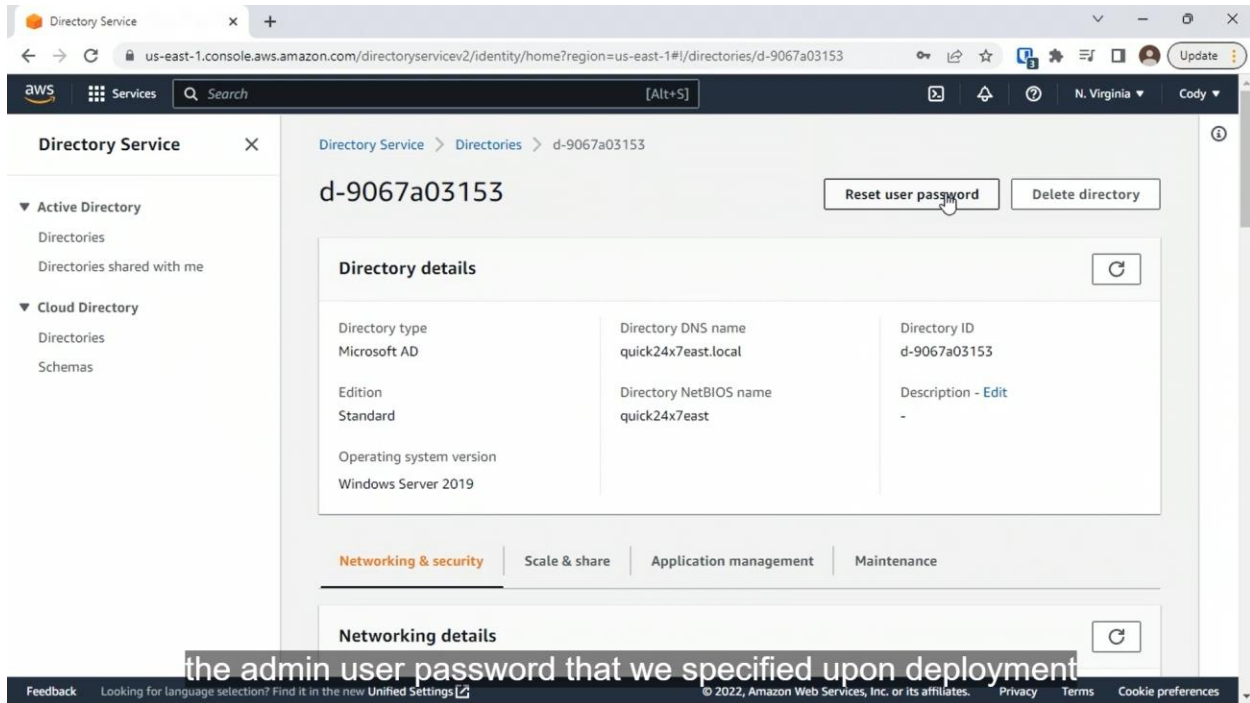
Select directory type [Info](#)

Directory types

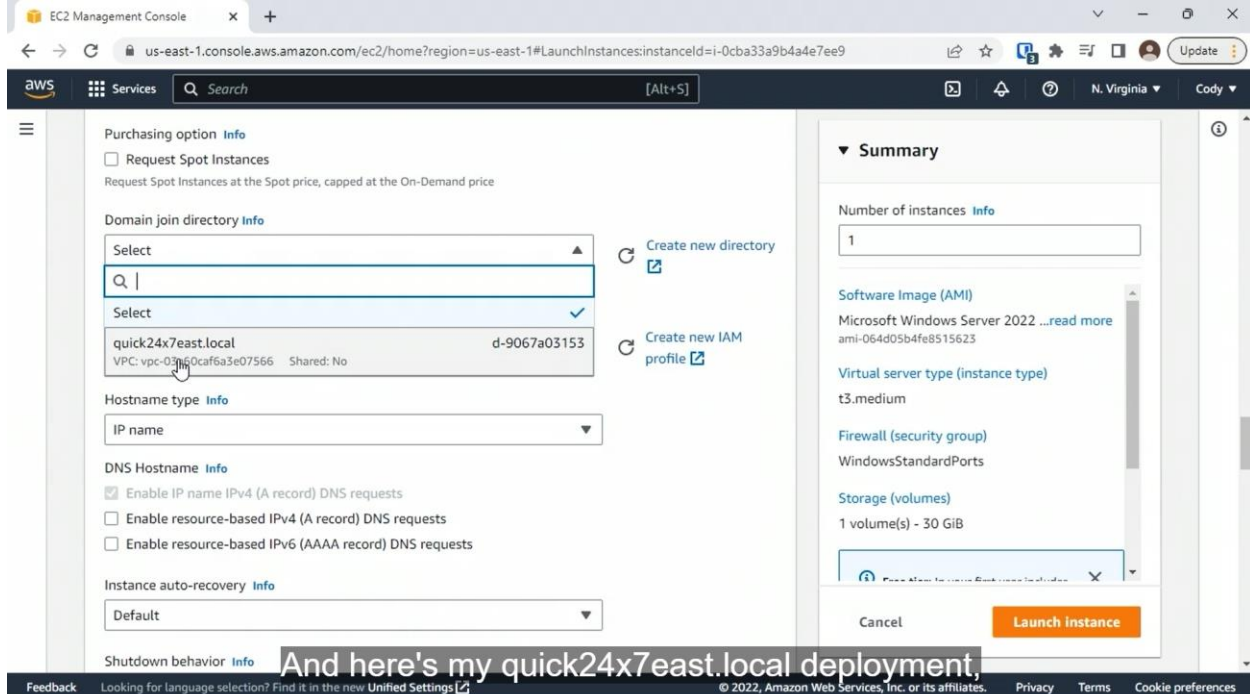
- ☐ AWS Managed Microsoft AD
- ☐ Simple AD
- ☐ AD Connector
- ☒ Amazon Cognito User Pools

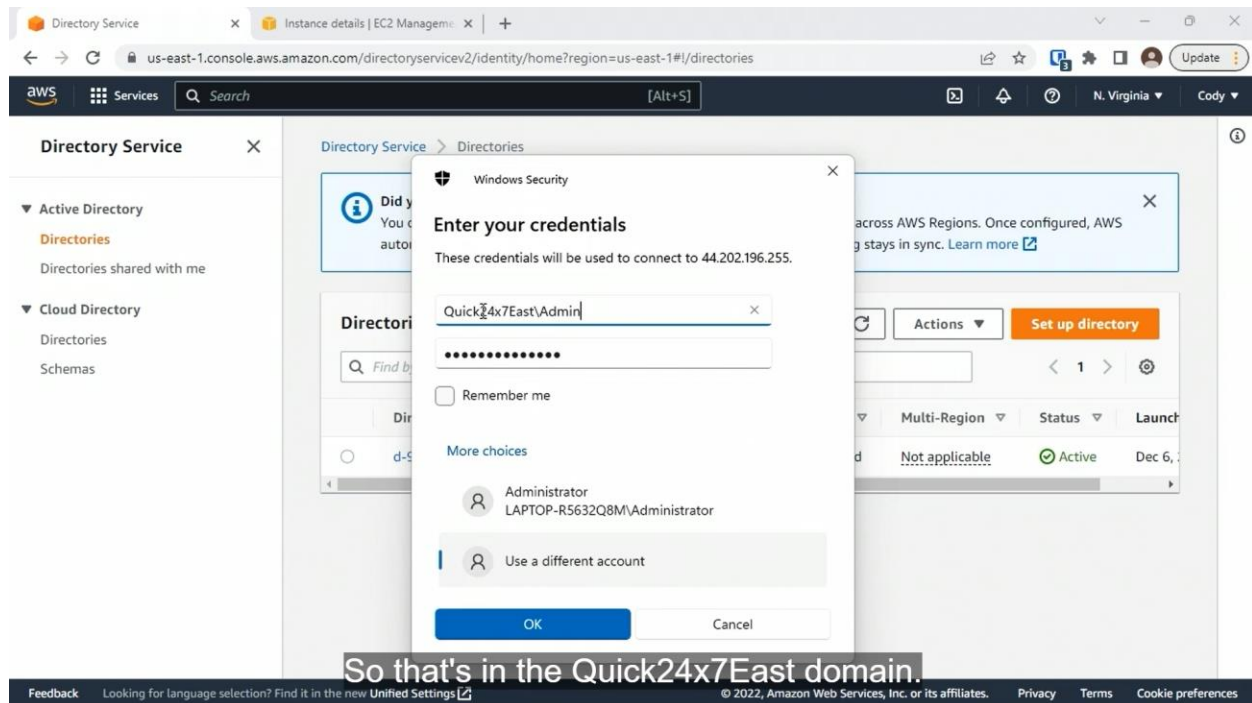
Amazon Cognito User Pools

With user pools, you can add user registration and sign-in features to your apps. Users can sign in with an email address, phone number, or user name rather than use an external identity provider like Facebook or Google. You can also create custom registration fields and store that metadata in your user directory. You can verify email addresses and phone numbers, recover passwords, and enable multi-factor authentication (MFA) with just a few lines of code.



Joining Stations to AWS Managed Active Directory (AD)





1. When are IAM roles used?
 - To assign permissions to a software or service entity
2. What is wrong with the following PowerShell expression?

Add-IAMUserToGroup -UserName MBishop
 - The group name is not specified
3. What is the default username for managed AD in AWS?
 - Admin
4. To which security best practice does the IAM Access Analyzer apply?
 - Least privilege
5. Which limitation applies to IAM group management?
 - Groups cannot be nested

6. Which details must be specified when creating a custom IAM policy?
 - Service
 - Actions

7. Which PowerShell cmdlet is used add IAM users to an IAM group?
 - Add-IAMUserToGroup

8. Which IAM resource consists of related resource permissions?
 - Policy

9. Which user access options are available when creating an IAM user in the console?
 - Console Access
 - Programmatic access

10. You are using the console to manage IAM users. After opening an IAM user account, where should you click to enable MFA?
 - Security credentials tab

11. How do Windows clients find domain controllers when joining an Active Directory domain?
 - DNS

12. Which CLI command is used to show the groups a user is a member of?
 - `aws iam list-groups-for-user`

13. To which AWS resources types can permissions boundaries be applied?
 - IAM users
 - IAM roles

14. Which CLI command is used to show the details of a specific IAM group?

- `aws iam get-group`

15. How do resource provider apps verify the validity of identity provider authentication tokens?

- They use the identity provider public key

16. You have noticed that the IAM Access Analyzer keeps reporting a finding related to an S3 bucket named BUCKET1 with public access enabled. BUCKET1 is used by a web app to serve product PDFs to website visitors. You no longer wish to see the finding related to BUCKET1. What should you do?

- Archive the finding for BUCKET1

17. Which options are available for configuring identity providers using the IAM Identity Center console?

- Managed AWS AD
- On-premises Active Directory